

BAILEY WALLACE

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Portfolio: baileywallace.ai • Company: lightspacelabs.com • Charlotte, NC • open to relocation

SUMMARY

Production photogrammetrist and geospatial engineer. My core work is stereo feature compilation in DAT/EM Summit Evolution with Civil 3D, and classifying county-scale LiDAR through a staged pipeline of TerraScan ground macros followed by an AI model I train and run. On top of that I build the internal software my team runs on, including a Python ArcGIS toolbox and a web dashboard now in daily use. Duke M.S. thesis training and validating deep-learning models on satellite imagery ($F1 = 0.933$, 95.9% pixel accuracy). Trained first as an environmental scientist, which shows up as care about validation, transferability, and stating plainly what an analysis does not support.

EXPERIENCE

ESP Associates, Photogrammetrist / GIS & AI Lead

Dec 2025 – Present • North Carolina

- Compile planimetric and topographic features in stereo from aerial imagery using DAT/EM Summit Evolution with Civil 3D. This is the department's core production work.
- Serve as the department's go-to GIS resource and AI subject-matter expert across production aerial-survey and LiDAR projects, setting how the team adopts machine learning in its geospatial pipelines.
- Produce and QC county-wide LiDAR and mapping deliverables in ArcGIS Pro, taking raw point clouds and Civil 3D / AutoCAD data through geoprocessing, classification, and QA/QC to deliverable-ready vector and raster outputs, including CAD-to-GIS conversion.
- Classify county-scale LiDAR point clouds through a staged production pipeline: Terrasolid TerraScan macros for the initial ground classification, then an AI model I train and run on that output, then manual editing and QC through to deliverable-ready data.
- Designed and shipped the internal web dashboard the team now uses daily to track editing progress across the production tile scheme: Python and Flask, SQLite, and a Leaflet map front end, with state-plane to WGS84 reprojection, progress rolled up by stage, person, and county, map annotation and flagging for review findings, GIS export, and a projected completion date computed from recent pace. Read-only against production data with audit logging.
- Built a Python ArcGIS toolbox that automates impervious-surface calculations for stormwater compliance, applying right-of-way logic across lot polygons and outputting a clean per-lot CSV; teammates run it independently, cutting hours per project and standardizing the deliverable.
- Translate technical geospatial output into guidance non-technical stakeholders can act on, including a department head who needs visibility without learning new software.

Duke University, Jim Zhang Lab, Graduate Researcher (M.S. Thesis)

2024 – 2026 • Durham, NC

- Built a two-stage deep-learning pipeline in Python and PyTorch on PlanetScope satellite imagery (3 m/pixel): a ConvNeXt Large classifier for detection and a DeepLabV3+ model with a ResNet-101 encoder for pixel-level segmentation.
- Constructed the ground-truth dataset directly, hand-annotating and spatially QC'ing 109 scenes; achieved $F1 = 0.933$ with perfect recall (1.000) on classification and 95.9% pixel accuracy on segmentation. Authored and defended the full thesis (March 18, 2026).
- Ran cross-location evaluation to quantify how accuracy transfers to a new geography without fine-tuning (IoU 0.448 to 0.319), publishing the degradation as a reusable template for model transferability rather than reporting validation-set wins alone.
- Ran the study as an international collaboration, coordinating with a research team at Mongolia National University alongside a Duke-based mentor, exchanging data and findings across a twelve-hour time difference.
- Modeled PM2.5 exposure following the Palisades wildfire, integrating satellite remote sensing, ground monitoring networks, and meteorological data into predictive surfaces of population exposure.

- Built and operate an AI-powered competitive-intelligence SaaS on the Anthropic Claude API, owning the full stack from data pipeline to customer-facing product.
- Built a multi-LLM evaluation loop that validates outputs for accuracy and consistency before they reach customers, plus the data and reporting layer that turns raw model output into decisions users act on.
- Manage all infrastructure (Vercel, Cloudflare Workers and D1, Railway, Postgres), tenant configuration, cost, and incident response.
- Designed and built the company website, and own marketing and positioning for the product.

SELECTED PROJECTS

Terra (*in build*). A three-agent satellite-imagery pipeline that turns raw imagery over a project bounding box into a verification report for construction lenders. Ingestion produces a GeoParquet of H3 hex cells with 20-dimensional feature vectors; downstream agents classify and reason over the cells to generate the deliverable. Replaces \$800 to \$2000 manual site inspections.

Lightsky (*live, lightsky.ai*). A team of AI assistants that runs the back office of a small business: nine specialist agents under an orchestrator covering inbox triage, lead capture, review response, SEO, marketing, site monitoring, and weekly reporting. Every agent drafts in the background and holds anything customer-facing until the owner approves, with per-agent permission scopes and auto-send unlocked task by task as trust builds. Next.js, Claude API, Railway, Postgres. Pivoted from an agent-observability SDK I built and ran, whose guardrail layer became the permission system.

SkySchool (*prototype*). Mobile-first PWA that teaches aerial photogrammetry and GIS from first principles, with hands-on demos for every concept. Vite, React, TypeScript.

EDUCATION

Duke University, M.S., Global Health

2026

Thesis: deep-learning detection and segmentation on PlanetScope satellite imagery. Coursework: GIS & Environmental Health, Predictive Modeling & Spatial Analysis.

UCLA, B.S., Marine Biology

2024

GPA: 3.7 / 4.0. Coursework: Geography / Environmental Studies, Global Health. NCAA Division I rowing.

TECHNICAL SKILLS

- **GIS & geospatial:** ArcGIS Pro, ArcGIS Online, ArcPy, ArcGIS ModelBuilder, QGIS, Global Mapper, MicroStation, Terrasolid TerraScan, DAT/EM Summit Evolution and DAT/EM production tools, geodatabase design and management, geoprocessing, spatial QA/QC, web maps and dashboards, cartography, coordinate systems and reprojection, CAD-to-GIS conversion, Civil 3D, AutoCAD, spatial databases.
- **Remote sensing:** PlanetScope multispectral imagery, LiDAR point clouds (LAS/LAZ), photogrammetry, raster and vector analysis, ground-truth annotation, accuracy assessment (F1, IoU, recall), domain-shift and cross-location evaluation, H3 hex grids, GeoParquet.
- **Machine learning:** PyTorch, ConvNeXt, DeepLabV3+, ResNet-101, image classification and semantic segmentation, training-set design, model validation and transferability analysis, random forests, regression, spatiotemporal and predictive modeling.
- **AI / LLM engineering:** Anthropic Claude API, AI-agent development, prompt engineering, AI vision, multi-LLM evaluation, guardrails and output validation, cost and usage monitoring.
- **Languages & data:** Python (GeoPandas, rasterio, Shapely, NumPy, Pandas, laspy), SQL, JavaScript / TypeScript, R, Jupyter, Postgres, SQLite, git and GitHub, command-line workflows.
- **Engineering & deployment:** Flask, Leaflet, React, Docker, Vercel, Cloudflare Workers and D1, Railway, Prisma, REST and JSON APIs, monitoring and incident response.